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ADAN 8888 - Applied Analytics Project - Spring 2026

Module 11 Assignment - Explain the Model, Analyze Risk, Bias and Ethical Considerations

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**HelpHerInvest: A machine learning-based investment recommendation system that aligns user interests and risk preferences with benchmark stock evaluation.**

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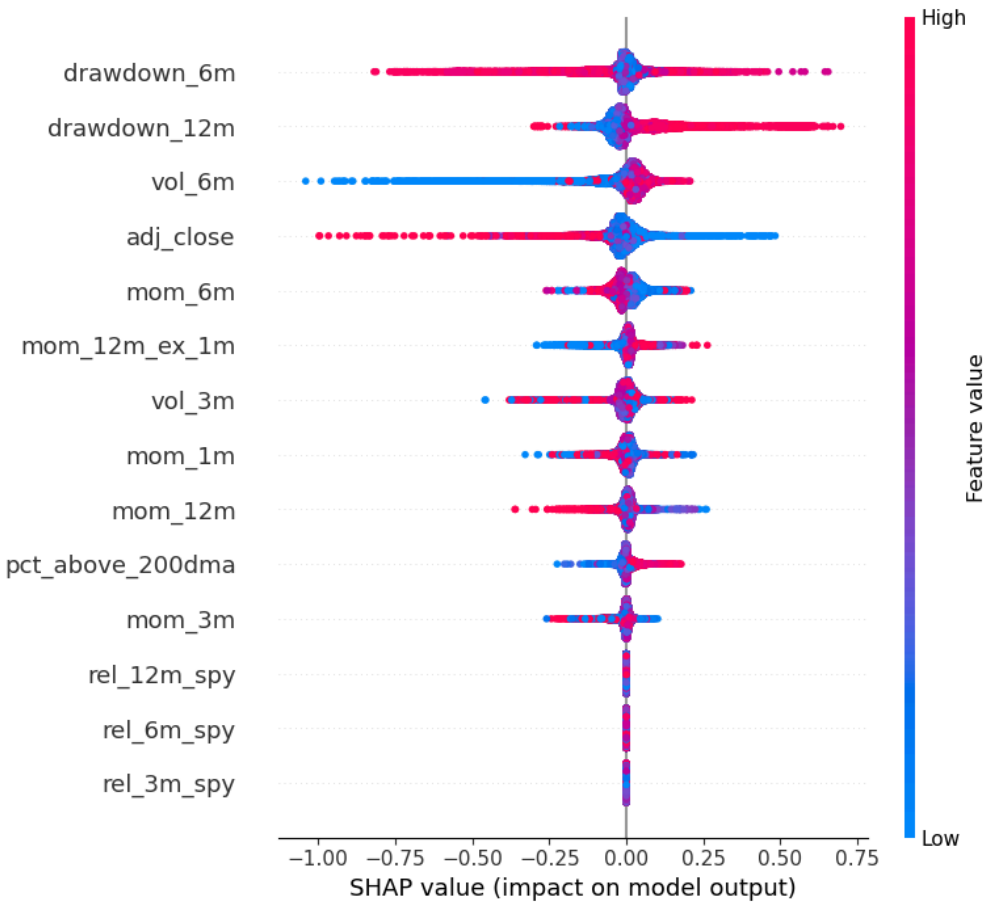
**Identifying the Important Features of the Model**

In order to help determine the most important features in our model, our group decided to utilize the SHAP (SHapley Additive exPlanations) package. SHAP is a model interoperability framework that is based on concepts of “game theory” and helps identify how much each feature contributed to the predicted result. Using SHAP can help identify the magnitude and direction (positive vs negative) each feature’s impact has on the predictions, which can be insightful when it comes to determining the appropriate features to use and whether certain features pose bias or risk.

This week's model and dataset selection was based on our week 10 assignment, where we were able to get the most favorable results by creating a cross sectionally ranked dataset and utilizing the winning Light Gradient Boosting Machine (LightGBM) model. When looking at the top five most important features of the LightGBM model, we calculated the absolute value of the SHAP values to determine the highest impact. Notably, features related to drawdown, momentum, and volatility were among the most impactful, suggesting nonlinear relationships within the data.

| feature | importance |
|---|---|
| drawdown_6m | 0.063273 |
| drawdown_12m | 0.060172 |
| vol_6m | 0.048394 |
| adj_close | 0.035371 |
| mom_6m | 0.027896 |

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### Select 5 Predictions at Random to Compare Results

After determining which features were most important, we wanted to review 5 predictions to compare whether the importance metrics were aligned. The most impactful features identified in the global importance analysis were present in each of the randomly selected predictions at least 3-4 times. One observation we had was that the vol_6m feature was in each of the random samplings' top 5 impactful features, which ranked third in the global importance analysis.

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--- Prediction 1 (Index 45705) ---
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| | feature | value | shap_impact |
|---|---|---|---|
| 11 | drawdown_12m | 0.635120 | 0.037848 |
| 2 | mom_6m | 0.298687 | 0.034795 |
| 9 | vol_6m | 0.181346 | -0.021282 |
| 1 | mom_3m | 0.297046 | 0.016213 |

| | | | |
|---|---|---|---|
| 8 | vol_3m | 0.237418 | -0.014500 |

**--- Prediction 2 (Index 37904) ---**

| | feature | value | shap_impact |
|---|---|---|---|
| 13 | adj_close | 9.790000 | 0.158641 |
| 1 | mom_3m | 0.993946 | -0.142808 |
| 9 | vol_6m | 0.996147 | -0.110166 |
| 11 | drawdown_12m | 0.003853 | -0.100295 |
| 4 | mom_12m_ex_1m | 0.002752 | -0.075638 |

**--- Prediction 3 (Index 19334) ---**

| | feature | value | shap_impact |
|---|---|---|---|
| 8 | vol_3m | 0.574614 | -0.037797 |
| 10 | drawdown_6m | 0.620354 | 0.027942 |
| 9 | vol_6m | 0.463122 | 0.017128 |
| 2 | mom_6m | 0.586335 | -0.017086 |
| 13 | adj_close | 33.192700 | -0.014584 |

**--- Prediction 4 (Index 18138) ---**

| | feature | value | shap_impact |
|---|---|---|---|
| 4 | mom_12m_ex_1m | 0.848764 | 0.020172 |
| 0 | mom_1m | 0.292697 | 0.019231 |
| 10 | drawdown_6m | 0.602070 | 0.017821 |
| 13 | adj_close | 280.663300 | -0.017176 |
| 9 | vol_6m | 0.270845 | 0.012419 |

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--- Prediction 5 (Index 17549) ---
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| | feature | value | shap_impact |
|---|---|---|---|
| 1 | mom_3m | 0.996550 | -0.145805 |
| 9 | vol_6m | 0.986199 | -0.094864 |
| 0 | mom_1m | 0.996550 | 0.086548 |
| 11 | drawdown_12m | 0.966072 | 0.067067 |
| 10 | drawdown_6m | 0.957734 | 0.050175 |

Overall, the SHAP values can be used to assess model risk and potential bias. Features with disproportionately high importance indicate areas where the model is highly sensitive, increasing the risk of instability if those inputs change. The SHAP analysis we performed revealed nonlinear relationships (which we had previously identified), suggesting that the model may behave differently across feature ranges or market regimes. By comparing SHAP values across subsets of the data, we can detect whether the model relies on different features under different conditions, which also may indicate bias or lack of generalization.

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**Protected Categories**

Our dataset does not appear to include direct protected attributes such as race, gender, age, religion, or disability status. Instead, our model is built around investment-related inputs, including stock performance measures, benchmark comparisons, and user preference signals such as interests and risk tolerance. Thus, we are not explicitly telling the model to make recommendations based on legally protected characteristics.

Even so, the absence of those variables doesn’t necessarily mean the model is free from fairness concerns. In our project, variables tied to user interests, financial behavior, or risk profile could indirectly reflect broader social patterns, such as differences in access to financial education or investment experience across groups. For this reason, protected categories are not directly used in the model, but fairness still matters because bias can enter through correlated inputs or through the structure of the training data itself.

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**Bias of the Model**

The main source of bias in our model comes from the data it was trained on, rather than from explicitly using protected characteristics. Because the model relies on historical stock performance features, it tends to favor patterns that have worked in the past. This creates a bias towards stocks that already show strong performance trends, which can limit the diversity of recommendations and overlook alternative investment strategies.

The choice of features also introduces bias. By focusing on momentum and volatility-based indicators, the model is more aligned with short-term or trend-following strategies and may underrepresent long-term or value-based approaches. If user risk preferences are included, the model may further reinforce bias by assigning users to simplified categories (low vs. high risk) and restricting recommendations based on those labels.

Lastly, there is also a representation issue within the dataset itself. If certain types of stocks appear more frequently in the data, the model may perform better for those categories and less reliably for others.

Overall, the bias in our project is not tied to protected attributes, but instead comes from historical market patterns, feature selection, and the structure of the dataset used to train the model. Features that require only small changes to significantly alter predictions highlight areas where the model may be overly sensitive, increasing prediction risk, but we did not find these were present within our dataset.

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**Bias Removal Strategies**

One way to reduce bias is to rethink which features we rely on most. Right now, a lot of our signals come from momentum and volatility, which naturally push the model toward stocks that have already been performing well. We could balance this by adding features that capture different investing styles, like longer-term stability or valuation-based signals.

Another step would be to look at how our dataset is distributed across different types of stocks. If certain sectors or market caps show up more often, the model will lean in that direction. Adjusting the dataset to better represent a wider range of stocks could make the recommendations more balanced.

**Stakeholder's Risks of Using the Model**

A big risk is that users may rely too heavily on the model without fully understanding how the recommendations are generated. Even if the model performs well overall, it could still be wrong in individual cases, especially when market conditions change.

Another concern is that the model may unintentionally narrow user choices. If the same types of stocks keep getting recommended, users might miss out on other opportunities simply because those options are not being surfaced. Of course, from a financial perspective, poor recommendations can have real consequences. And there is also risk tied to changing market conditions. The model is trained on past data, so if the market shifts, the patterns it learned may no longer hold. Without regular updates, the quality of recommendations can decline over time.

On a broader and ethical level, if users act on these recommendations, it's not always clear where accountability falls - whether on the model, the developers, or the user. Because of these risks, the model should be positioned as a support tool rather than a decision-maker. It can guide users, but it should not replace independent judgment or basic understanding of investing.